

Test before Selection (2024) Full marks: 100 Time: 2 Hours

"The battlefield of programming is not against others, but against the limits of your own mind. Every line of code, every algorithm conquered, brings you closer to victory."

MCQ:

20x1=20

1. The access specifier that provide access only to its child class -

- a) private b) public
c)protected d) default

2. char a= (char) ('A'+32) , a=?

- [illegible]

3. The type of inheritance which is not supported in java

- a) Multilevel b) Multiple
c) Hierarchical d) Hybrid

4) `System.out.println(Math.sqrt(1-9));` (it's mathematically impossible, but it prints something)

- a) -3.0 b) 3.0
c) error d) NaN

5) "CONCATENATION".indexOf('T')

- a) 5 b) 9
- c) System.exit(0) d) break

6) Which one is in correct order of precedence

- a) ++ ! / - > b) ! / ++ - >
c) / - ++ ! > d) / + ! ++ -

7. “ARUSH”.compareTo(“AROSH”)

1. 5
2. 6
3. -6
4. -5

8. $k = 3 > 4 ? 5 : 2 < 3 ? 6 : 8;$

- a) $k=5$ b) $k=6$
c) $k=8$ d) $k=3$

9. `int k= a[a.length]`

- a) Syntax Error
- b) Logical Error
- c) Runtime error
- d) No Error

10. Write the possible range of values for the expression `Math.random() * 10;`

- a) 0 to ≤ 10
- b) 0 to < 10
- b) 1 to ≤ 10
- d) 1 to < 10

11. total physical size of a 2D char array of order 3x4

- a) 24 bytes
- b) 48 bytes
- c) 12 bytes
- d) 96 bytes

12. `System.out.println(Math.round(4.3));`

- a) 4
- b) 5
- c) 4.0
- d) 5.0

13) Correct method to check whether a character is letter or digits is-

- a) `Character.isDigitOrLetter(ch)`
- b) `Character.isLetterOrDigit(ch)`
- b) `ch.isLetter()`
- d) `ch.isLetterOrDigit()`

14) calling a method by passing object is known as

- a) Call By Value
- b) Call by Reference
- c) Call by object
- d) Call by default

15) Method that changes state of an object

- a) Friend Function
- b) pure function
- c) inner function
- c) impure function

16) `Integer.valueOf()` returns

- a) Integer object
- b) data of type int
- b) String
- d) character

17. `String s[]= {"Apple", "Guava", "Strawberry", "Pomegranate "}; int k= s[2].length(); k=?`

- a) 5
- b) 10
- b) 11
- d) 9

18) int a[][]= { {2,3}, {4,5}, {5,6} }; int k= a[1][1] + a[2][1];

- a) 11 b) 10
c) 22 c) 8

19. System.out.println(1/2+ "\t"+ 1.0/2.0 + "\t" + "YO");

- a) 0.5 0.5 YO b) 0 0.5 yo
b) 0 0.5 YO d) 1 0.5 YO

20. extension of byte code created by JAVA compiler is

- a) .java b) .class
b) .file d) .binary

SAQ. Answer the following

10x2=20

1. System.out.println(2+3);

System.out.println("Sum="+2+3);

System.out.println("Sum"= + (2+3));

2. double p= Math.max(Math.round(2.5), Math.ceil(2.2));
p=?

3. find output if (i) choice= 'a' (ii) choice='x'

```
switch(choice){  
    case 'b': System.out.println("No");  
        break;  
    case 'a': System.out.println("That is");  
    case 'c': System.out.println("Correct !");  
        break;  
    default: System.out.println("Wrong Input");  
}
```

4. Convert the if-else construct to ternary

```
if(x==4){  
    v= v-0;  
}else if( x<4){  
    v= v+1;  
}else{  
    v=v  
}
```

5. char ch1= 'Z';

int n= ch1-32;

Sytem.out.println(n);

System.out.println((char)n);

6. Convert the if else statement to switch case

```
if(x==3){
System.out.println("React");
}else if(y==4){
System.out.println("Express");
}else{
System.out.println("Solidity");
}
```

7. Write prototype of a function that takes a String and returns an integer

8. Convert to while() and do while()

```
int a=10;
for(int i=1; i< a/2; i++){
a+=I;
}
```

9. for (int i = 1; i <= 10; i++) {

if (i * i > 30) {

break;

}

System.out.print(i + " ");

}

10. for (int i = 1; i <= 3; i++) {

int j = 1;

while (j <= 2) {

System.out.print(i + " " + j + " ");

j++;

}

} Find output and number of times the loop executes.

Section B

Solve any four:

15x4=60

1. Design a class with the following specifications:

Class name: Student

Member variables:

- name – name of student
- age – age of student
- mks – marks obtained
- stream – stream allocated

(Declare the variables using appropriate data types.)

Member methods:

1. Student() – constructor to initialize data members with their default values
2. void accept():
Accept name, age, and marks using methods of the Scanner class.

3. void allocation():

Allocate the stream as per the following criteria:

- mks $\geq$ 300: Science and Computer
- mks $\geq$ 200 and $<$ 300: Commerce and Computer
- mks $\geq$ 75 and $<$ 200: Arts and Animation
- mks $<$ 75: Try Again

4. void print():

Display student name, age, mks, and stream allocated.

Call all the above methods in the main method using an object.

2. Define a class to accept values into an array of double data type of size 20 in ascending order. Accept a double value from the user and search in the array using the Binary search method. If the value is found, display the message "Found" with its position where it is present in the array. Otherwise, display the message "not found".

3. Take a String from user and replace all vowels with @ and all consonants with #

4. Define a class to overload the function print as follows:

void print()

54321

4321

321

21

1

void print(int n)

To check whether the number is a lead number. A lead number is the one whose sum of even digits is equal to the sum of odd digits.

e.g. 3669

Odd digits sum = $3 + 9 = 12$

Even digits sum = $6 + 6 = 12$

3669 is a lead number.

5. Take a Matrix of 4x4 size from user. check if it's diagonal array or not. If sum of left diagonal (primary) and sum of right diagonal (secondary) is equal, It is called a diagonal array

6. Define a class to accept values in integer array of size 10. Find sum of one digit number and sum of two digit numbers entered. Display them separately. Example: Input: a[] = {2, 12, 4, 9, 18, 25, 3, 32, 20, 1}

Output: Sum of one digit numbers: $2 + 4 + 9 + 3 + 1 = 19$

Sum of two digit numbers: $12 + 18 + 25 + 32 + 20 = 107$